

Highly Branched Copolymers with Degradable Bridges for Antifouling Coatings

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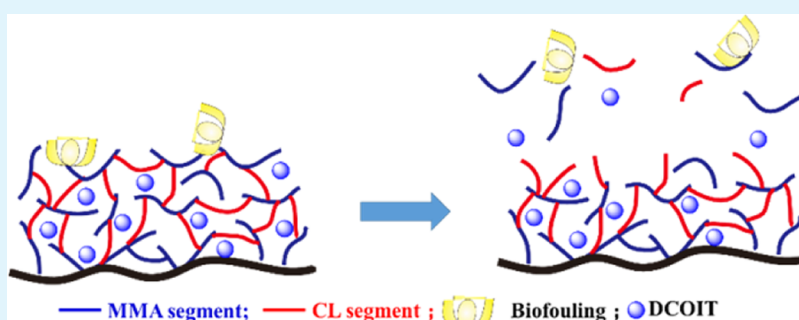
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ABSTRACT: The antifouling properties of traditional self-polishing marine antifouling coatings are mainly achieved based on their hydrolysis-sensitive side groups or the degradable polymer main chains. Here, we prepared a highly branched copolymer for self-polishing antifouling coatings, in which the primary polymer chains are bridged by degradable fragments (poly- ϵ -caprolactone, PCL). Owing to the partial or complete degradation of PCL fragments, the remaining coating on the surface can be broken down and eroded by seawater. Finally, the polymeric surface is self-polished and self-renewed. The designed highly branched copolymers were successfully prepared by reversible complexation mediated polymerization (RCMP), and their primary main chains had an M_n of approximately $3410 \text{ g} \cdot \text{mol}^{-1}$. The hydrolytic degradation results showed that the degradation of the copolymer was controlled, and the degradation rate increased with increasing contents of degradable fragments. The algae settlement assay tests indicated that the copolymer itself has some antibiofouling ability. Moreover, the copolymer can serve as a controlled release matrix for antifoulant 4,5-dichloro-2-octylisothiazolone (DCOIT), and the release rate increases with the contents of degradable fragments. The marine field tests confirmed that these copolymer-based coatings exhibited excellent antibiofouling ability for more than 3 months. The current copolymer is derived from commonly used monomers and an easily conducted polymerization method. Hence, we believe this method may offer innovative insights into marine antifouling applications.

KEYWORDS: antifouling coatings, highly branched polymers, degradable polymer, reversible complexation mediated polymerization, dynamic surface

INTRODUCTION

Biofouling is a constant and substantial problem in the marine industry because the adhesion of undesirable marine fouling organisms on immersed underwater structures, such as pipelines, ship hulls, and bridge pillars, drastically increases the frictional resistance of the structure and causes significant fuel consumption.^{1–3} Therefore, numerous antifouling strategies including chemical, physical, and biological methods have been developed.^{4–8} Among these approaches, employing antifouling coatings is the most effective and versatile method.

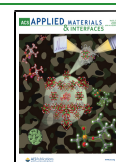
As early as the mid-nineteenth century, the first antifouling coatings appeared, containing copper, arsenic, or mercury oxide as toxicants dispersed in linseed oil or shellac.⁹ After this, the use of tributyltin (TBT)-based antifouling coatings was popular for a long period due to their high efficiency in preventing the settlement and growth of marine organisms.

Coatings of this type are based on an insoluble acrylic copolymer with antifouling side groups bonded to the main polymer chain by ester bonds (Scheme 1a). After immersion in seawater, the carboxyl side groups (carboxyl-TBT) bond of the copolymer undergo controlled hydrolysis, making the copolymer film brittle and easily erodible in seawater, thus exposing a new fresh area of the coating. However, these coatings were subsequently found to have disastrous effects on the marine

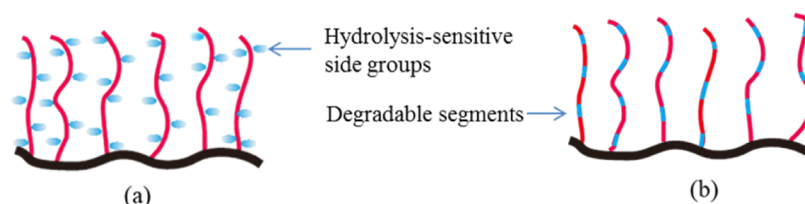
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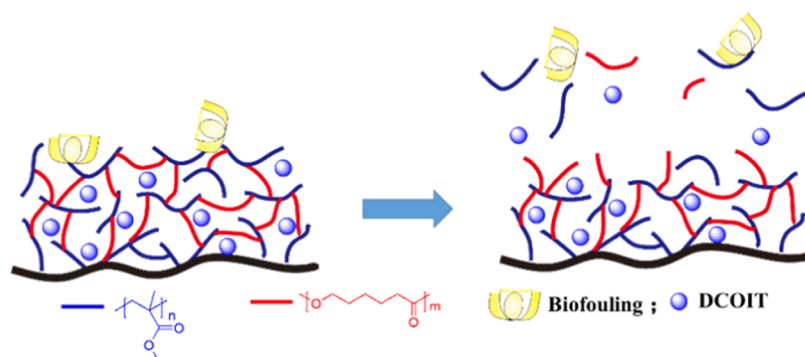
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Scheme 1. Self-Polishing Antifouling Polymers Based on (a) Hydrolysis-Sensitive Side Groups or (b) Degradable Main Chains



Scheme 2. Self-Polishing Antifouling Mechanism in the Present Study



environment and have been completely banned since January 1, 2008. Since then, developing more environmentally friendly antifouling coatings has become an urgent topic for both researchers and coating manufacturers.^{10–13}

Aliphatic polyesters are typical biodegradable polymers that can achieve a renewable surface via the cleavage of polymer main chains. More importantly, they can be assimilated without any harmful effects on the environment, making them promising candidates for antibiofouling coatings (Scheme 1b).^{9,14–19} Among the families of biodegradable aliphatic polyesters, ϵ -caprolactone (CL) copolymers are particularly attractive because of their good degradability and compatibility.²⁰ However, the availability of the cyclic monomer used for copolymerization is quite limited, significantly restricting diversifications in the coating products and limiting their extensive application range. In addition, besides the chemical composition, polymer topography is also important for the application. Branched polymers exhibit greater solubility and lower viscosity compared with their linear analogues and are suitable candidates for coatings.²¹ However, only star-shaped degradable polymers are applied for marine antifouling coating.²² This is because most of the synthesized methods for the degradable branched polymer are labor-intensive and time-consuming.^{23,24} Exploiting an easily conducted synthesized method for degradable branched polymers from commercial monomers remains a challenge. The present study conceives a new strategy for degradable branched polymers and aims to find a facile and easily conducted method for the preparation of self-polishing antifouling coatings based on conventional vinyl monomers and lactone.

RESULTS AND DISCUSSION

Antifouling Mechanism and Polymer Resin Synthesis. (Methyl) acrylate resins have been widely used for self-polishing antifouling coatings. However, their main chains are composed of C–C covalent bonds, making them hard to degrade. The antifouling effect for this type of resin was

achieved based on the hydrolysis of side groups.^{25–27} In contrast, the main chains of CL copolymer resins can be easily degraded by the dissociation of ester groups. However, these resins are hard to functionalize due to their limited monomer resources. Finding an easy method that combines the advantages of both resins for antifouling coatings is the main aim of the present work. For this purpose, hybrid copolymerization of vinyl monomers and lactones seems to be a feasible method.^{28–33} However, the high price of the catalysts has currently hindered their mass production. Here, we designed a type of highly branched copolymer used for self-polishing antifouling coatings, in which the primary chains are composed of methyl methacrylate (MMA) fragments, and the bridges linking two poly(methylmethacrylate) (PMMA) chains are poly- ϵ -caprolactone (PCL) fragments (Scheme 2). The self-polishing effect is mainly achieved depending on the degradation of PCL bridges by seawater, which results in the complete dissociation of the highly branched copolymer to ultimately renew the surface. Moreover, PMMA is one of the most commonly used polyacrylate that is widely used in coatings owing to its remarkable film formation performance and adhesion to various substrates. Note that the remaining undegradable PMMA segments might form particles in the sea. However, these segments have low toxicity and are easily metabolized.³⁴

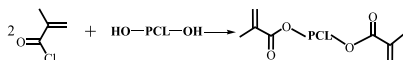
Before the synthesis of the predesigned highly branched polymers, three key factors need to be considered. First and foremost, the length of the primary chain cannot be too long; otherwise, even if the PCL fragment bridges are completely degraded, the remaining PMMA chains in the matrix still cannot be removed from the surface, resulting in a poor antifouling effect. This phenomenon implies that only “living” polymerization technologies should be considered. In addition, the length of the PCL fragment bridges should be controlled because long-chain PCL is crystalline, which decreases its degradability in seawater. Finally, the method employed for the synthesis of the highly branched polymers must be simple, and

the polymerization conditions should be easily achieved, as these factors determine the competitiveness of the product.

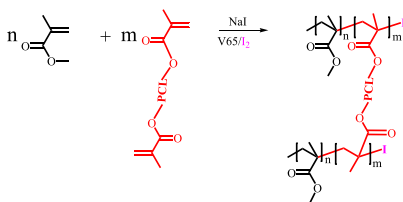
Based on the above consideration, RCMP is employed in the current work.^{35–39} This method is quite easy to handle and only requires the introduction of iodine (I_2) and sodium iodide (NaI) in traditional polymerization. The mechanism of RCMP is shown in Figure S1, and the synthesis route for a highly branched copolymer is shown in Scheme 3. First, vinyl-

Scheme 3. Simplified Mechanism of the Formation of Highly Branched Copolymers via Reversible Complexation Mediated Polymerization (RCMP)

(1) Synthesis of vinyl-functionalized PCL (ν -PCL).



(2) Synthesis of highly branched copolymers with degradable bridges.



functionalized PCL (ν -PCL) was prepared by the reaction between PCL and methacryloyl chloride. To decrease the crystallinity, the degree of polymerization (DP) of PCL is controlled to below 15. Here, the actual molecular weight of the synthesized PCL calculated from the 1H NMR spectrum (Figure S2) or size-exclusion chromatography (SEC) is 1.54×10^3 and 2.34×10^3 g·mol^{−1}, respectively. Then, MMA and ν -PCL were copolymerized by RCMP with the addition of iodine (I_2), sodium iodide (NaI), and 2,2'-azobis (2,4-dimethyl valeronitrile) (V65). The length of the primary chain could be easily adjusted by the molar ratio of $[MMA]/[I_2]$.

Characterization of Highly Branched Copolymers.

Figure 1 shows the 1H NMR spectrum of the synthesized

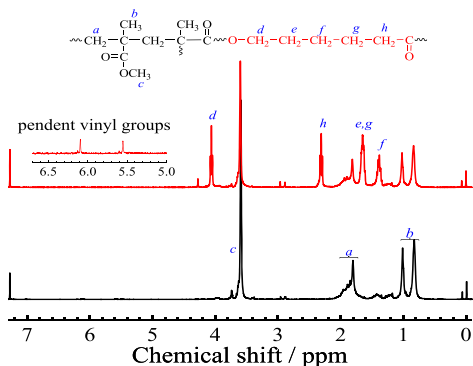


Figure 1. 1H NMR spectra for P(MMA-CL_{0.33}) and *l*-PMMA.

highly branched copolymer with a molar ratio of $[MMA]/[\nu\text{-PCL}]/[V65]/[I_2]/[NaI]$ of 40:1.0:1.0:0.4:0.4. For comparison, linear PMMA was also prepared by free radical polymerization. As shown, in addition, to the signals for the protons of MMA units at 1.62–2.06 (a), 0.67–1.27 (b), and 3.45–3.79 (c) ppm, peaks at 3.96–4.41 (d), 1.54–1.76 (e, g), and 1.29–1.47 (f) ppm, respectively, due to the CL units were

clearly observed. Besides, faint signals at 5.73–6.50 ppm attributed to the pendent acrylate vinyl groups in ν -PCL were also observed. It should be noted that the pendent vinyl groups cannot contribute to the formation of branched polymer because only reactions with both the double bonds in ν -PCL could give a branched structure. An integrals comparison between these signals to those of CL revealed that 94.5% of the vinyl groups in ν -PCL were consumed, suggesting that a large majority of ν -PCL successfully reacted to produce branched structures. This result can be further verified by the Fourier transform infrared (FT-IR) spectra in Figure S3. In addition, the CL content in the copolymer can be easily calculated according to the 1H NMR spectrum. Here, the weight ratio of MMA to CL units is 67:33, and hence, the resulting copolymer was named P(MMA-CL_{0.33}).

The resulting copolymer was further analyzed using SEC. As shown in Table 1, its absolute weight-average molecular weight (M_w) reached 1.21×10^5 g·mol^{−1} and the polydispersity ($\bar{D}$) was 8.33. This high molecular weight and wide distribution were due to its highly branched structures and suggested success in the synthesis of the target copolymer. In addition, the intrinsic viscosity ($[\eta]$) plots of P(MMA-CL_{0.33}) in Figure 2 are all below those of *l*-PMMA. It should be noted that the low viscosity of the copolymer facilitates its coating ability, and hence, this property is favorable because the low viscosity means that a low amount of organic solvent is needed for this process.^{42,43} Besides, the Mark–Houwink parameter α can be used to describe the shape and compactness of a polymer. For a linear coil in a good solvent, $0.5 < \alpha < 0.8$. For a branched polymer, α ranges from 0.3 to 0.5.^{40,41} Here, the calculated α value is 0.48, further confirming the branched structure of the resulting copolymer.

Figure 3 shows the DSC curves for ν -PCL and P(MMA-CL_{0.33}). For ν -PCL, a crystal temperature (T_c) of 47.03 °C was clearly observed. However, when ν -PCL was copolymerized with MMA, no T_c was found in the temperature range from −50 to 120 °C. The high crystallinity of PCL causes this polymer to have poor film-forming properties and low degradability in seawater.^{20,44} The current results seem to suggest that the crystallinity of PCL was significantly restricted after copolymerization, which endows the copolymer with good degradability, thus improving its suitability as a self-polishing antifouling coating.

Antifouling Properties. As mentioned above, the antifouling effect of the present copolymer mainly arises from the dynamic surface formed by the degradation of the PCL fragment bridges. After partial or complete degradation of PCL fragments, the remaining PMMA fragments are easily eroded by seawater due to their short chains. Therefore, the polymeric surface is renewed, and antifouling is achieved. To confirm this deduction, the degradation of the copolymer was investigated.

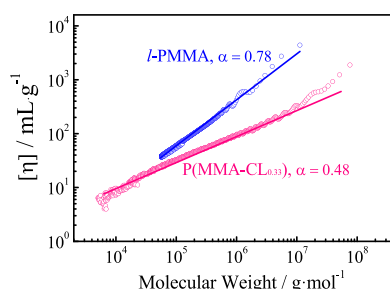
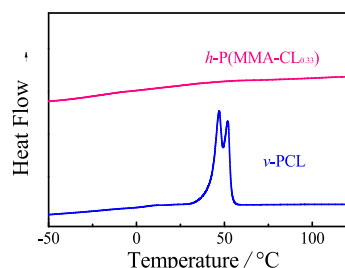
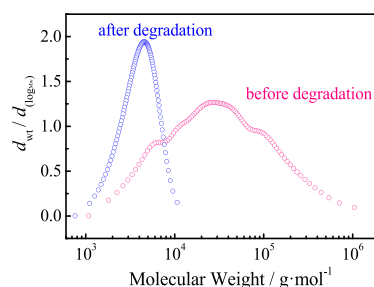
Figure 4 shows the evolution of the different molecular weight distributions for P(MMA-CL_{0.33}) before and after degradation in a NaOH aqueous solution. Before the degradation of the copolymer, its M_w reached 1.21×10^5 g·mol^{−1}, and the curve of the copolymer showed a broad multimodal distribution. After the degradation of the copolymer, its curve became a narrow monomodal peak, and its M_n was only 3.41×10^3 g·mol^{−1}. More importantly, it is difficult for the remaining component after degrading to form a film (Figure S4), and it is easily dispersed in water, which endows the copolymer surface with good self-polishing ability.

Table 1. Preparation of Highly Branched Copolymers^a

sample	[MMA]/[DV ^b]/[V65] [I ₂]/[NaI]	DV	F _{CL} (wt %)	M _n (10 ³ g·mol ⁻¹)	M _w ^c (10 ⁴ g·mol ⁻¹)	Đ	a ^d
ν-PCL ^e				2.34			
h-PMMA	40:0.8:1.0:0.4:0.4	BDDMA ^f		12.9	5.35	4.15	0.44
P(MMA-CL _{0.33})	40:1.0:1.0:0.4:0.4	ν-PCL	33	14.5	12.1	8.33	0.48
P(MMA-CL _{0.39})	40:1.2:1.0:0.4:0.4	ν-PCL	39	13.2	9.22	6.99	0.46
P(MMA-CL _{0.27})	40:0.8:1.0:0.4:0.4	ν-PCL	27	9.84	3.34	3.34	0.50

^aThe polymerization for branched copolymers was conducted at 70 °C. ^bBranched agents containing two vinyl groups. ^cDetected from MALLS.

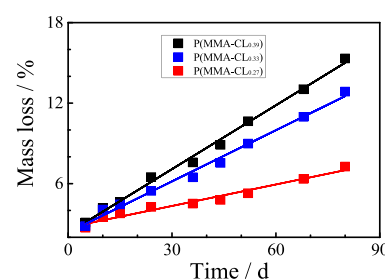
^dMark–Houwink parameter. ^eVinyl-functionalized PCL. ^f1,4-Butanediol dimethylacrylate.

Figure 2. Mark–Houwink curves for P(MMA-CL_{0.33}) and h-PMMA.Figure 3. DSC curves for P(MMA-CL_{0.33}) and ν-PCL.Figure 4. Evolution of the different-molecular-weight distributions for P(MMA-CL_{0.33}) before and after degradation.

Inspired by this positive result, we then prepared two additional copolymers with different CL contents, P(MMA-CL_{0.27}) and P(MMA-CL_{0.39}). The SEC analysis shows that each copolymer has a high molecular weight and a wide distribution (Table 1 and Figure S5). Moreover, no T_c was observed for all copolymers, suggesting that their crystallinities were completely destroyed. Generally, employing more difunctional monomer would help to obtain the polymer with a high molecular weight. However, the observed M_w of P(MMA-CL_{0.39}) is lower than that of P(MMA-CL_{0.33}) (in Table 1). This might be attributed to the formation of a slight amount of microgel, which would be removed during the sample preparation process for the SEC analysis. It should be noted that the copolymer h-PMMA synthesized using a short-chain branched agent (1,4-butanediol dimethylacrylate,

BDDMA) has difficulty forming a film. However, all of the samples bridged with PCL fragments have good film performance, which makes them much more suitable for coating than PCL homopolymer or branched PMMA.

The self-polishing ability of the three special synthetic samples was first studied by analyzing their degradability in artificial seawater (ASW) at 25 °C. As seen in Figure 5, the

Figure 5. Time-dependent mass loss of P(MMA-CL_{0.27}) (red), P(MMA-CL_{0.33}) (blue), and P(MMA-CL_{0.39}) (black) in ASW.

mass losses of copolymers are almost linear with time over the entire 3 months, indicating a constant degradation rate. In addition, Figure 5 shows that the degradation rate of the copolymers increased with increasing CL content, which can be due to the increasing ester group density in the copolymer. This result confirms that the mass-loss rate could be efficiently controlled by adjusting the CL contents, which is significant for the design of marine antibiofouling coatings. It should be noted that each copolymer for the present study exhibits a very broad distribution due to the co-existing many species with different molecular weights, which also have an effect on the weight loss during the hydrolytic degradation testing.

For further studying the antifouling property of the prepared highly branched copolymer, the algicidal adhesion of the copolymers is examined by observing the growth of phaeodactylum tricornutum, which is widespread in the Xiamen Sea. Generally, phaeodactylum tricornutum can grow on coating surfaces. Here, we expected that these algae could be easily removed by renewing the coating surface due to its degradability. The algicidal adhesion results are shown in Figure 6. As seen, close to 60% of the control panel is covered by phaeodactylum tricornutum. After coating with copolymers on the surface of the panel, the amount of phaeodactylum tricornutum that adhered decreased from 24.45 to 4.74% with increasing CL content from 27 to 39%. These results suggest that the synthesized highly branched copolymer has good antialgae property, and a high amount of PCL in the copolymer matrix helps to increase antifouling performance. In other words, the dynamic surface formed by the degradation of the PCL fragment bridges contributes to good antifouling ability. In addition, it should be noted that the experiments

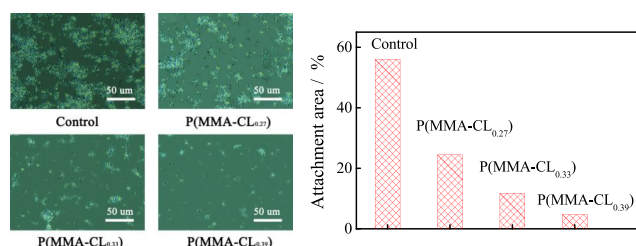


Figure 6. (left) Image and (right) attachment area of phaeodactylum tricornutum settled on the surface of the coated copolymers with different CL contents for 14 days.

were conducted under a closed system, where the acid produced from the degradation of the PCL segments would also have a negative effect on the growth of alga.⁴⁵

Generally, the degradable polymer itself would have a good antifouling performance if its degradation rate is enough fast.⁵ However, a high degradation rate leads to inevitably short service life. Therefore, it can be a good strategy to improve the antifouling duration of a degradable polymer by adding an antifoulant. Here, eco-friendly organic antifoulant 4,5-dichloro-2-octyl-isothiazolone (DCOIT) was selected since it has been granted by the European Union (EU) and the US Environmental Protection Agency (EPA) due to its rapid degradation in marine environments.

Figure 7 shows the cumulative release of DCOIT from the copolymers under static conditions in ASW at 25 °C. Each film

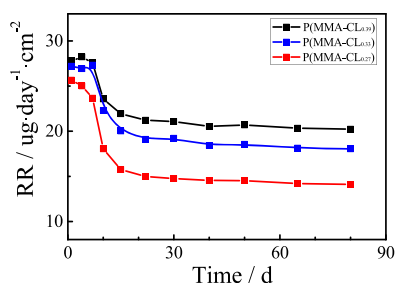


Figure 7. Release rate (RR) of 4,5-dichloro-2-octyl-isothiazolone (DCOIT) from P(MMA-CL_{0.27}) (red), P(MMA-CL_{0.33}) (blue), and P(MMA-CL_{0.39}) (black) in ASW at 25 °C.

has a higher release of DCOIT during the initial 10 days than at the later stages. This is not surprising because the dispersion of DCOIT on the surface is much easier than the dispersion of the DCOIT blended in the matrix. After the initial period, the release rate of each sample reached a constant value, suggesting that DCOIT is uniformly dispersed on the polymer and that the copolymers degrade at a constant rate. This is very important for the performance of an antibiofouling system because the fast release of DCOIT shortens the service life of the system; in contrast, the slow release of DCOIT results in poor antifouling efficiency. Only a controlled release of DCOIT would endow the coating with good antifouling properties. Moreover, Figure 7 also shows that the release rate of DCOIT increased with increasing CL content in the matrix, which is in accordance with the degradation test.

To further examine the antifouling efficiency, the long-term antifouling performance of the system was evaluated by respectively covering the three samples with 10 wt % DCOIT on epoxy surfaces for marine field tests. Figure 8 shows that the control panel with epoxy is completely fouled

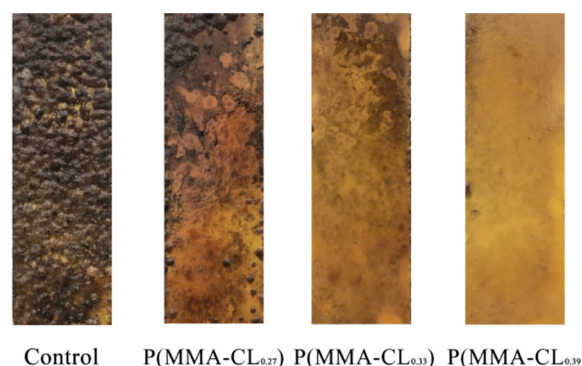


Figure 8. Images of panels coated with the prepared copolymers with different CL contents based coatings containing 10 wt % DCOIT after immersion in seawater for 3 months in Xiamen Bay in China.

by marine organisms. However, for the panel coated with the copolymer/DCOIT composites, much less fouling is observed. Moreover, the fouling on the surface decreased with increasing CL content, indicating that the antifouling ability improved with increasing CL content. We deduce two reasons for the good performance of the present antifouling coating. First, the degradation of the PCL bridges produces a continuous release of DCOIT, making a good resistance to macro- and microfouling organisms. Moreover, the degradation of the PCL bridges and the hydrolysis of PMMA segments might form a changing dynamic surface, which also helps to form an antifouling surface. This mechanism was proposed by Zhang et al. as “dynamic surface antifouling (DSA)”.⁵

CONCLUSIONS

In summary, we have prepared a highly branched copolymer for self-polishing antifouling coatings in which the polymeric primary chains are bridged with PCL fragments. Due to the degradation of bridges, the remaining coating surface is broken and self-renewed. Three predesigned highly branched copolymers with different CL contents were successfully synthesized by RCMP. Their hydrolytic degradation results showed that the degradation of the copolymer is controlled, and the degradation rate increases with increasing contents of degradable PCL fragments. The copolymer itself has some antialgae ability. Furthermore, the copolymer has a controlled release rate of DCOIT. Marine field tests confirmed the excellent antibiofouling ability of the copolymer-based coatings.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at <https://pubs.acs.org/doi/10.1021/acsami.9b22748>.

Experimental section, mechanism, additional ¹H NMR spectra, FT-IR spectra, and film images; mechanism of the reversible complexation mediated polymerization (RCMP); and images for the films coated using the degraded copolymer before (a) and after (b) immersing in ASW (PDF)

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Notes

The authors declare no competing financial interest.

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